2027 MSA Strategic Initiatives — CALIBER Proposal (Draft)

Working draft for a proposal to the MSA Strategic Initiatives program, 2027 call. Bracketed `[TODO]` items need lab input before submission. Citations are drawn from the Edison research reports in `outputs/msa_2027_strategic_initiatives/`.

Call constraints (2027 cycle)

- **Deadline:** 11:59 PM Eastern Time, November 1, 2026
- **Funding:** suggested level up to $15,000 for a one-year project or $20,000 total over two years; significantly higher possible for exceptional projects with major benefits to the Society; 3–4 awards expected
- **Eligibility:** primary applicant must be a current MSA member; detailed budget required
- **Scope:** bold, innovative *new* programs; not intended to sustain existing programs or the primary research/educational missions of institutions
- **Primary areas of emphasis targeted** (of the six listed in the call):
    - *Area 2:* MSA-branded development and dissemination of information and open source tools related to advances in microscopy, microanalysis, data management, and image analysis
    - *Area 6:* Collection, preservation, management, and dissemination of microscopy-related information, including technical procedures, basic theories, scientific backgrounds, MSA archived material, and related resources
- **Secondary areas touched:** student involvement (Area 1), member benefits (Area 3), academic/governmental/industrial partnerships (Area 5)

Title

The MSA Microanalysis Parameter Commons: MSA-branded open-source tools and a preserved knowledge base for selecting high-fidelity acquisition parameters, built on CALIBER

[Alternate: "CALIBER for the Community: An MSA-Branded Open Platform and Knowledge Commons for Microanalysis Acquisition Parameters"]

Applicant team

- **Principal applicant:** `[TODO: name — must be a current MSA member; confirm membership status]`
- **Co-applicants / advisors:** `[TODO: draw from the reviewer/collaborator shortlist — see issue 10]`
- **Host lab:** Vertical Cloud Lab `[TODO: institution details]`

Requested support

$20,000 over two years *(scale to $15,000 / one year by dropping Year 2 items if preferred)*

Summary

Quantitative microscopy lives or dies by acquisition parameters. Physical standards can hold SEM-EDS to roughly ±5% relative uncertainty while standardless analysis can err by ±30%, with normalization hiding the damage [1]; raising EBSD collection speed from 54 to 154 patterns per second dropped correct indexing from 34.9% to 9.3% [2]; and supplying a 12-phase candidate list instead of the correct two caused *zero* patterns to index correctly despite ideal beam settings [3]. Our own LPBF AlSi10Mg work mirrors this: standardless EDS over-reported trace Mg several-fold, and indexing yield swung from 40% to 80% on beam-current and phase-list corrections alone. The rationale behind "recommended" settings — why standards demand overvoltage ≥ 1.8, where defaults break down — is scattered across decades of literature, standards whose derivations are rarely documented, vendor-proprietary software, and the memory of senior microscopists; when they retire, it leaves with them [1,15].

We propose an MSA-branded **Microanalysis Parameter Commons**: (1) an open-source release of **CALIBER**, a retrieval-augmented, uncertainty-aware recommender that suggests literature- and physics-grounded SEM, EDS, EBSD, and XRF acquisition parameters for a stated sample and analysis goal, and (2) a curated, versioned, citable knowledge base that collects and preserves *why* parameters are chosen — technical procedures, underlying physics, and the derivations behind standard settings — cross-linked to MSA's own archives, where much of this knowledge already sits unorganized [12,13]. An open benchmark campaign on novel alloys fills a documented evidence gap (no controlled factorial acquisition studies exist for LPBF AlSi10Mg [1,3]), students earn stipends curating entries under expert review, and results flow through M&M tutorials, a Microscopy Today article, and MSA's web presence. The award funds a community resource, not our lab's research program.

Alignment with the 2027 areas of emphasis

Call area	How this initiative addresses it
Area 2 — MSA-branded open source tools (primary)	CALIBER's RAG pipeline, parameter recommender, and benchmark tooling released under a permissive license as an MSA-branded resource for microscopy, microanalysis, data management, and image analysis
Area 6 — collection, preservation, dissemination of microscopy knowledge (primary)	The Parameter Commons captures technical procedures, basic theory, and the scientific background behind acquisition settings in versioned, DOI-citable records, cross-linked to MSA archived material (M&M proceedings, Microscopy Today)
Area 1 — student involvement	Paid student curation corps; curation is a structured on-ramp into quantitative microanalysis
Area 3 — member benefits	Members get a practical tool that shortens time-to-good-data on unfamiliar samples; contributor recognition for members

Call area	How this initiative addresses it
Area 5 — partnerships	Academic and national-lab partners contribute validation datasets and expert review `[TODO: name partners]`

Background and need

- **Parameters determine fidelity, quantitatively.** Standards-based SEM-EDS can reach about ±5% relative uncertainty; standardless can reach ±30%, and normalized totals conceal errors [1]. EBSD throughput, frame averaging, voltage, binning, and above all phase-list selection each swing indexing success by tens of percentage points — one-frame patterns were almost entirely misclassified, and an over-inclusive 12-phase list produced zero correct assignments [2,3,4]. In XRF of Al alloys, minor Mg has been over-reported by 40% relative and trace Mg by 5000%, while matrix-matched calibration holds major elements below 5% error [5,6,7].
- **The "why" is sparsely documented and hard to access.** Standards such as ISO 22309 and ASTM E1508 supply operating thresholds (e.g., overvoltage ≥ 1.8, take-off near 35°) but not the derivations, uncertainty surfaces, or validation data needed to adapt them to a specific detector, alloy, and goal; investigators comparing two standardless systems could not even determine why their accuracies differed, because the algorithms and calibration databases were vendor-constrained [1]. Protocol repositories and methods sections preserve the procedural *what* but not the expert *why* — alternatives considered, failure modes, stopping criteria [8,11,14]. FAIR metadata schemas structure acquisition context but by design do not capture the causal rationale for choosing it [9,10]. And the practical expertise behind parameter choices demonstrably leaves science when experienced practitioners retire or move on [15].
- **Open tools stop short of guidance.** HyperSpy, kikuchipy, pyxem, EMsoft, DREAM.3D, ImageJ/Fiji, and napari are post-acquisition analyzers, simulators, or viewers; none recommends acquisition settings, grounds advice in citable literature, or quantifies its own uncertainty — and autonomous-microscopy work to date is task-specific and policy-driven rather than literature-grounded [8,11,13]. CALIBER is connective infrastructure over this ecosystem, not a duplicate of it.
- **The validation data don't exist yet.** No controlled, ground-truthed factorial study links acquisition settings to chemical and crystallographic fidelity for LPBF AlSi10Mg — published settings are examples, not validated optima [1,3,4]. The open benchmark in this initiative creates exactly that dataset, for the community.
- **Much of the source knowledge is already MSA's.** The strongest parameter studies and ecosystem papers appear in *Microscopy and Microanalysis* and M&M proceedings [3,12,13] — MSA archived material that the Commons would organize, cross-link, and make actionable (Area 6).

Objectives and deliverables

1. **O1 — MSA-branded open-source CALIBER release.** Public repository (permissive license) containing the retrieval-augmented recommender, uncertainty-aware feedback loop, and corpus-building tools; MSA branding and hosting arrangement agreed with Council; built to interoperate with the existing open ecosystem (HyperSpy/kikuchipy data models, NeXus/NXem metadata [9]) rather than replace it. *Deliverables: v0.1 (Y1Q2), v1.0 (Y2Q2); documentation and install-free web demo.*

2. **O2 — The Microanalysis Parameter Commons.** A public, versioned knowledge base of parameter-rationale records (per modality × parameter × material class: the recommended range, the physics behind it, the primary sources, and known failure modes), with DOIs per release, a contribution/review workflow, and cross-links into MSA archives (M&M proceedings, Microscopy Today). *Deliverables: schema + 50 seed records (Y1Q2), 150+ records spanning SEM, EDS, EBSD, XRF (Y2Q4).*
3. **O3 — Open validation benchmark.** A published benchmark on novel alloy samples (laser powder bed fusion Al-Si and the broader alloy campaign) that fills the documented absence of controlled factorial acquisition studies [1,3]: raw spectra/patterns archived and re-analyzed offline outside vendor software, so recommended vs. naive parameters can be compared reproducibly; early-stopping quality heuristics (e.g., EBSD confidence-index screening after partial scans) evaluated and reported. *Deliverables: open dataset with FAIR metadata per MaRDA recommendations [9] (Y2Q1); benchmark report.*
4. **O4 — Dissemination and training.** M&M tutorial/workshop, a Microscopy Today article, and a student curation corps designed around what makes trainee curation sustainable: bounded tasks, structured mentorship and review, version control, and visible credit on DOI'd releases [15,16,17]. *Deliverables: workshop at M&M `[2027 and/or 2028]`; 6–10 student curators trained.*

Work plan and timeline (two years)

Period	Milestones
Y1 Q1	MSA branding/hosting agreement; Commons record schema; seed extraction for the highest-impact parameters (EDS: accelerating voltage + `[TODO: second parameter under evaluation]`; EBSD: beam current, voltage, phase list; SEM imaging baseline set)
Y1 Q2	CALIBER v0.1 open-source release; Commons alpha with 50 seed records; student curator recruitment
Y1 Q3	Validation campaign 1 (LPBF Al-Si): archived raw data, offline re-analysis, uncertainty-aware feedback loop on recommended settings
Y1 Q4	Public beta; M&M tutorial submission; Microscopy Today article draft
Y2 Q1	Benchmark dataset release (FAIR metadata, DOI); community contribution workflow opens
Y2 Q2	CALIBER v1.0; XRF modality coverage; 100+ records
Y2 Q3	M&M workshop; assessment against success metrics
Y2 Q4	150+ records; sustainability handoff to `[TODO: MSA committee/FIG]`; final report to Council

Budget (draft — $20,000 over two years)

Item	Amount
Student curation and development micro-stipends (6–10 curators × 2 yr)	$8,000
Instrument time for the open validation benchmark (SEM/EDS/EBSD/XRF sessions)	$6,000
Hosting, compute, and DOI/archiving costs (2 yr)	$2,000

Item	Amount
M&M tutorial/workshop materials and costs	$2,500
Dissemination (open-access fees, Microscopy Today piece)	$1,500
Total	**$20,000**

`[TODO: adjust to institutional rates; confirm no overhead applies; detailed justification per MSA form]`

Benefit to MSA and success metrics

- Positions MSA as the home of *preserved*, *citable* microanalysis practice — knowledge that currently evaporates with retirements [15] — under its own brand (Areas 2 and 6), following the model of societies that built trusted open resources (rOpenSci, MolSSI, ELIXIR) [16,18].
- A concrete, recurring member benefit: shorter time-to-reliable-data on unfamiliar samples.
- Metrics: number of published Commons records (target 150+); tool users/downloads and web-demo sessions; benchmark dataset downloads; workshop attendance; student curators trained; records cross-linked to MSA archived material; new/retained members reporting use `[TODO: baseline survey mechanism]`.

Sustainability

Year 2 transitions governance to an MSA body `[TODO: identify committee or Focused Interest Group]` with a lightweight editor-plus-reviewers model. The precedent literature is specific about what makes society-branded resources survive — governance by a trusted organization, integration with existing community activities (here: M&M, Microscopy Today), contributor recognition, and maintenance capacity — and about the failure mode to avoid: volunteer-only efforts detached from established infrastructure [16,18]. Post-award hosting costs are minimal (static site + repository); curation continues through the student pipeline and member contributions credited on DOI'd releases.

Team, partners, and reviewers

`[TODO: PI and co-applicants; partner labs and agencies (Area 5) — candidates from the byu-vcl shortlist; note the call excludes microscopy vendors as partners]`

References

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Appendix: supporting research artifacts

Full Edison reports backing this draft, in `outputs/msa_2027_strategic_initiatives/`:

- `need_evidence_answer.md` — quantitative evidence report with a proposal-ready needs statement; includes the evidence table (`need_evidence_artifact-00.md`)
- `deep_landscape_answer.md` — landscape review with ten distilled gap statements (`deep_landscape_artifact-00.md`) and a tool-by-tool gap table (`deep_landscape_artifact-01.md`)